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5 August 1985
SUPERSEDING
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# MILITARY SPECIFICATION

# LOADING STANDARDS, LIQUID TRANSFER, BULK:

# LIQUID FUELS, STATIONARY, 3-INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers stationary loading standards for transfer of bulk liquid petroleum fuels.

1.2 Classification. The loading standards shall be of the following types, as specified (see 6.2):

Type I - Loading standard, liquid transfer, bulk: stationary, 3-inch, rigid assembly.
Type II - Loading standard, liquid transfer, bulk: stationary, 3-inch, sectional assembly.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| W-C-440 | - Clip, Electrical, General Specification for. |
| W-R-550 | - Rod, Ground (With Attachments). |
| QQ-S-635 | - Steel Plate, Carbon. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| QQ-S-781 | - Strapping, Steel and Seals. |
| WW-T-700/6 | - Tube, Aluminum Alloy, Drawn, Seamless, 6061. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-C-843 | - Cushioning Material; Cellulosic. |
| PPP-T-97 | - Tape, Pressure-Sensitive Adhesive, Filament Reinforced. |

MILITARY

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|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-F-3541 | - Fittings, Lubrication. |
| MIL-G-3859 | - Grease Gun, Hand Operated, Lever, Push and Screw Type. |
| MIL-A-7021 | - Asbestos Sheet, Compressed, for Fuel, Lubricant, Coolant, Water, and High Temperature Resistant Gaskets. |
| MIL-A-8625 | - Anodic Coatings, for Aluminum and Aluminum Alloys. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-V-12003 | - Valves, Plug: Cast-Iron or Steel, Manually Operated. |
| MIL-G-20241 | - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive. |
| MIL-A-22895 | - Adhesive, Metal Identification Plate. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-R-24049 | - Rope, Polypropylene. |
| MIL-C-27487 | - Coupling Halves, Quick-Disconnect, Cam-Locking Type. |
| MIL-S-81733 | - Sealing and Coating Compound, Corrosion Inhibitive. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw-Thread Standards for Federal Services. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |



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| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment and Facilities. |
| MS27029 | - Coupling Half, Quick Disconnect Cam-Locking Type Plug, Dust, Type X. |
| MS27030 | - Gasket-Coupling Half, Quick-Disconnecting, Cam-Locking Type. |

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B16.5 - Steel Pipe Flanges and Flanged Fittings.
ANSI B16.9 - Wrought Steel Buttwelding Fittings (ISO R 285).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

TRUCK TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA RP28 - Companion Flange.

(Application for copies should be addressed to the Truck Trailer Manufacturers Association, 1020 Princess Street, Alexandria, VA 22314

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 193 - Alloy Steel Bolting Materials for High-Temperature Service.
ASTM A 312 - Seamless and Welded Austenetic Stainless Steel Pipe.
ASTM B 210 - Aluminum-Alloy Drawn Seamless Tubes.
ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)



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AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS D1.1 - Structural Welding Code, Steel.
ANSI/AWS D1.2 - Structural Welding Code, Aluminum.
D10.7 - Aluminum Pipe, Gas Shielded-Arc Welding.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The loading standards shall be as shown on figures 1 and 2 and as specified herein. Each loading standard shall dispense fuel at a working pressure of 50 pounds per square inch gage (psig).

3.2 Pressure requirement. The portion of the loading standards between the fuel inlet and the loading valve shall withstand a hydrostatic proof pressure of 150 psig, for not less than 2 minutes, without leakage or permanent deformation or other defects that could harmfully affect the performance and serviceability of the loading standards.

3.3 Human factors. The loading standard shall conform to the controls and ground workspace design requirements of MIL-STD-1472.

3.4 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3, 6.2, and 6.3).

3.5 Standard product. The loading standards furnished under this specification shall be the manufacturer's current standard product, except for those modifications or additional changes required in that standard product in order to comply with the requirements of this specification.

3.6 Materials. Materials shall be as specified herein and as shown on figure 1 and figure 2. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.6.6). Materials shall be compatible with all hydrocarbon base (petroleum and synthetic) and alcohol base fuels as well as mixtures of the two. Fuel-wetted parts shall not be fabricated of copper, brass, or bronze materials.

3.6.1 Aluminum alloy. Aluminum alloy shall conform to ASTM B 210, Alloy 6061-T6.

3.6.2 Stainless steel. Stainless steel shall conform to ASTM A 312.



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3.6.3 Material deterioration and control. The loading standards shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the loading standards may be exposed.

3.6.4 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.6.5 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.6.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the loading standard may be newly fabricated from recovered materials to the maximum extent practicable, provided the loading standard produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the loading standard.

3.7 Buttwelding fittings. Buttwelding fittings shall conform to ANSI B16.9.

3.8 Flanges. Flanges shall be class 150 and shall conform to the requirements specified in ANSI B16.5 and TTMA-RP28; all flanges shall be furnished with the necessary gaskets, bolts, nuts, and washers.

3.9 Gaskets. A gasket shall be furnished for each pair of companion flanges and shall be fabricated of material conforming to MIL-A-7021, class 1, 1/16-inch thick.

3.10 Components.

3.10.1 Loading arm assembly. The loading arm assembly shall be a 3-inch diameter commerical type, equipped with O-ring seals and having the following operational characteristics:

a. Swing freely through 360 degrees in a horizontal plane.
b. Swing freely through a vertical arc from not less than 30 degrees below to 70 degrees above the horizontal plane.
c. When manually lowered to the loading position, the loading arm shall automatically remain in that position during loading. When released from loading position, the loading arm shall be returned to its original position by adjustable counterweights.



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d. When swung through its vertical arc, the drop tube shall hang in a vertical position by its own weight.

3.10.1.1 Loading valve. The loading valve shall be a lever operated, self-closing type, of such construction as to assure a smooth shockproof closure and shall be equipped with a vacuum breaker. The valve shall function without failure and shall not leak or show any indication of structural failure when subjected to a hydrostatic proof pressure of 150 psig, for not less than 2 minutes.

3.10.1.2 Remote-control handle. A remote-control handle shall be connected to the loading valve lever by means of a metal connecting rod as shown in figure 2; or when specified (see 6.2), a rope shall be attached to the lever and extending to the handle or eye provided on the single swing joint as shown on figure 1. The rope shall be polypropylene, conforming to MIL-R-24049, type I, 3/8 inch diameter, 12 feet long.

3.10.2 Sliding tube. The sliding tube shall be an assembly of two telescoping tubes. The outer tube shall be seamless, drawn, round aluminum tubing conforming to WW-T-700/6, type I, temper T6. The inner tube shall be aluminum or 300 series stainless steel with an inside diameter of not less than 3 inches. The tube shall be fully guided and shall slide easily from approximately the 5-foot telescoped position to the 9-foot fully extended position. One end of the assembly shall be externally threaded. The other end shall have TTMA RP28 flange welded on. The outer tube shall have a stuffing box with a packing compatible with petroleum fuels. The stuffing box shall be equipped with a packing gland and a lugged and threaded take-up ring for adjustment. The sliding tube assembly including the drop tube, single swing joint, and the outside tube stuffing box, shall not leak when subjected to a 25 psig hydrostatic proof pressure for 5 minutes.

3.10.3 Swivel joint, double. The double swivel joint shall be equipped with double rows of ball bearings for each rotative direction. The bearings shall be sealed to prevent leakage and entry of fuels into the bearing cavities, and shall be provided with lubrication fittings specified in 3.11.1. One end of the joint shall have a TTMA RP28 flange welded on, the other-inlet-end shall have class 150 flange. The double swivel joint shall provide a 360-degree rotation in both the horizontal and the vertical planes.

3.10.4 Swivel joint, single. The single swivel joint shall consist of two aluminum 90-degree elbows, grooved on the inside, to receive O-ring seals, and on the outside to receive a split retainer collar to prevent disjointing. Elbows and O-ring seals shall ride on a sleeve, acting as a trunnion. A handle shall be cast integrally with the elbow which is to receive the drop tube. The inlet and outlet shall be internally threaded. The single swivel joint shall provide a 360-degree rotation in one plane only.

3.10.5 Valves, plug. The plug valves shall conform to MIL-V-12003, type I, class 150, 4-inch size. Pattern shall be venturi or regular opening. Valve ends, inlet and outlet shall be flanged, with class 150 flange.



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3.10.6 Grounding reel. The grounding reel shall be a commercial, enclosed type, and shall have the following characteristics and component parts:

a. Steel cable 3/32-inch, zinc coated, approximately 40 feet in length and a minimum tensile strength rating of 1,000 pounds. The cable shall terminate in an electric power clip with lug conforming to W-C-440, type PCL, nominal current rating, 100 amperes.
b. Automatic rewind for cable.
c. Provisions for vertical mounting on the support bracket.
d. Ratchet device to latch the cable at intervals.
e. Instruction plate specified in 3.13.1.

3.10.7 Rod, ground. The ground rod shall be a driven sectional rod conforming to W-R-550, type III, class B.

3.10.8 Ground connections. A double-terminal ground clamp shall be placed around the riser pipe, approximately as shown on figures 1 and 2. Ground wire for terminals shall be 0.24 inch nominal diameter, No. 4 AWC, stranded copper wire. One 10-foot length and one 5-foot length of ground wire shall be furnished.

3.10.9 Base. The base and gusset plates shall be made from steel plate conforming to QQ-S-635, composition SAE 1020, dimensions as shown on figures 1 and 2. The plates shall be welded in place at the end of the standpipe.

3.10.10 Coupling half, inlet connection. The coupling half shall be flanged female cam-locking type quick-disconnect, size 4 inches, equal to OPW 633-LBS coupler, and shall be equipped with a gasket conforming to MS27030-9 and dustplug with chain attachment conforming to MS27029-17. "OPW" refers to the OPW Division of the Dover Corporation.

3.11 Lubrication. All surfaces requiring lubrication shall be provided with means for lubrication.

3.11.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I or III, as applicable. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 without removal or adjustment of accessories or parts. Fittings shall incorporate provisions for release of excess lubricant to the atmosphere.

3.11.2 Lubricant. Grease lubrication, including lubrication of sealed bearings, shall be with standard military thermally stable greases conforming to MIL-G-10924 or MIL-G-23827, as applicable.

3.12 Safety requirements.

3.12.1 Static electricity protection. With the loading standard fully erected and ready for operation, and the cable from the grounding reel extended to its full length, there shall be not more than 3 ohms resistance between the tip of the drop tube and the grounding cable attachment clip.



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3.12.2 Electric conductivity. Where electric conductivity for static grounding is required, protective coatings and any materials with insulating properties shall be removed from between contacting parts. The flange type coupling, swing joint, and loading arm assembly shall be provided with graphited packing and gaskets, sliding brushes, or other positive means of insuring the electrical conductivity required.

3.12.2.1 Support bracket. The area of the support bracket which comes in contact with the grounding reel shall not be painted or anodized to insure a positive ground.

3.13 Identification marking. The loading standards shall be identified in accordance with MIL-STD-130. The marking shall be applied to the loading standards on plates conforming to MIL-P-514, type III, composition C, of type I, class 1, grade B material, thickness 0.008 to 0.012 inch; black background with aluminum lettering. Plate shall be bonded to the riser pipe, in a suitable location, with adhesive conforming to MIL-A-22895, class B.

3.13.1 Instruction plates. The loading standards shall be equipped with instruction plates suitably located, describing any special or important procedure to be followed in operating and servicing the loading standards. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, class 1, material thickness 0.008 to 0.012 inch; black background with aluminum lettering. Plates shall be bonded to the riser pipe, in a suitable location, with adhesive conforming to MIL-A-22895, class B. One plate shall be located at the grounding reel in a conspicuous location. This plate shall give the following warning: "Copper grounding clip must be attached to the truck prior to starting loading operations".

3.14 Treatment and painting. The portions of the loading standards normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G. Unless otherwise specified (see 6.2), the finish paint shall be semigloss, color No. 24084.

3.14.1 Anodic coating. The loading standards and component parts, fabricated from aluminum alloy, when not painted, shall be anodized in accordance with MIL-A-8625, type II, class 2. Color shall be as specified.

3.15 Interchangeability. All identical components of each type loading standard shall be interchangeable.

3.16 Threaded connections. All threaded connections shall conform to FED-STD-H28 and shall be assembled by the contractor. The threaded connections except drop tube shall withstand a hydrostatic pressure of 150 psi for 2 minutes without any evidence of leakage.

3.17 Types.

3.17.1 Type I, rigid. The type I, rigid loading standard shall be as shown on figure 1 and as specified herein. Eight hook anchor bolts 5/8 inch diameter



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by 12 inches long with nuts and washers conforming to ASTM A 193 shall be furnished for anchoring of the base plate to a concrete footing. The concrete shall not be furnished.

3.17.1.1 Coupling half, can-fill. The coupling half shall conform to MIL-C-27487, type III, style B, class 1, size 2 inches, and shall be equipped with a gasket and dust cap with chain attachment conforming to MIL-C-27487, type IX, class 1, size 2-inch. A nipple, threaded to fit the 2-inch male coupling half, shall be welded in place to the 4-inch riser pipe in location as shown on figure 1, at 5 feet above ground level. A hook to hold the can-fill nozzle shall be welded to the riser pipe in the approximate location shown on figure 1.

3.17.1.2 Drop tube. The drop tube shall be aluminum tubing conforming to WW-T-700/6, type I, temper T6. The tube shall be not less than 3 inches inside diameter by 5 feet long. One end of the tube shall be externally threaded for connecting to the single swivel joint. The other end shall be beveled as shown on figure 1; or when specified (see 6.2), shall be fitted with a T-deflector similar to the one specified for the type II loading standard.

3.17.2 Type II, sectional. The type II, sectional loading standard shall be as shown on figure 2, and as specified herein. The concrete shall not be furnished.

3.17.2.1 Riser pipe. The riser pipe shall be in four sections as shown in figure 2, three 6-foot sections and one 5-foot section. When specified (see 6.2), the third (upper) 6-foot section shall not be furnished. This section of the riser pipe is not required when the loading standard is to be used for loading of tank trucks only.

3.17.2.2 Drop tubes. The drop tubes shall be aluminum tubing conforming to WW-T-700/6, type I, temper T6. Inside diameter shall be not less than 3 inches. Two lengths of tube shall be furnished. One tube shall be 5 feet and the other 11 feet in length. Each drop tube shall be equipped on one end with the female half coupling equal to OPW 128-R ring and OPW 132-T tail. The male half coupling shall be equal to OPW 132 and shall be connected to the single swivel joint. The opposite end of each drop tube shall be equipped with an anti-splash deflector equal to OPW 464 deflector. "OPW" refers to the OPW Division of the Dover Corporation. When specified (see 6.2), the 11-foot drop tube shall not be furnished (see 6.1.2).

3.18 Workmanship. The loading standards shall be free from defects such as misalined components, incomplete welds, cracks, burns, leaks, and other defects that could impair the operation and serviceability of the loading standards. All parts shall be clean and free from dirt, sand, grease, oil, and metal chips. All external surfaces shall be free of burrs, sharp edges and corners, except where sharp edges or corners are required and not detrimental to safety.

3.18.1 Welders and welding.



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3.18.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Steel.
ANSI D1.2, Structural Welding Code, Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes. This certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.18.1.2 Welding (aluminum). Welding performed in the fabrication of the loading standards shall be in accordance with AWS D10.7, Aluminum Pipe, Gas Shielded Arc Welding. The surface of parts to be welded shall be free from grit, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.18.1.3 Castings and forgings. All parts, components, and assemblies of the loading standards which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all section 3 and 5 requirements. The inspections in section 4 are the minimum to be used to demonstrate compliance. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and figures, as applicable.

4.1.3 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection procedures specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. When a first article sample is required, the first article sample shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. When a first article sample is required, the first article sample shall be tested as specified in 4.5.2. Failure of any test shall be cause for performance of the inspection specified in 4.1.3.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a unit of product shall be all individual parts and components required for assembling one complete loading standard.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105, level S-2, for both the sliding tube assembly and the complete loading standard.

4.4.3 Examination. Samples selected in accordance with 4.4.2.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 defects per hundred units for major and 6.5 defects per hundred units for minor.



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4.4.4 Tests.

4.4.4.1 Sliding tube assembly. Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2.3.2. AQL shall be 4.0 percent defective.

4.4.4.2 Loading standard (complete). Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3.1. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The loading standards shall be examined for the following defects:

Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Components missing or not as specified.
104. Sliding tube assembly not as specified.
105. Adjustable assemblies that cannot be adjusted or do not perform the function intended.
106. Grounding reel and component parts not as specified.
107. Positive means for electric conductivity not provided.
108. Identical components not interchangeable.
109. Not lubricated where required.
110. Workmanship not as specified.
111. Loading valve not as specified.
112. Remote-control handle not as specified.
113. Swivel joints, double and single not as specified.
114. Pressure requirements not as specified.
115. Human factors not as specified.
116. Buttwelding fittings not as specified.
117. Flanges not as specified.
118. Gaskets not as specified.
119. Valves, plug not as specified.
120. Ground rod not as specified.
121. Ground connections not as specified.
122. Base not as specified.
123. Coupling half, inlet connection not as specified.
124. Safety requirements not as specified.
125. Threaded connections not as specified.
126. Type I or type II standard not as specified.
127. Welders certification not as specified.
128. Welding not as specified.
129. Castings and forgings not as specified.
130. Grounding reel not as specified.
131. A minimum tensile strength rating of 1,000 pounds not as specified.



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132. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
133. Contractor does not have documentation available for identification of material, material finishes or treatments.
134. Aluminum alloy not as specified.
135. Stainless steel not as specified.
136. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the loading standard.

Minor

201. Lubrication fittings missing or not as specified.
202. Identification marking missing, incomplete, or illegible.
203. Instruction plates missing, incomplete, or illegible.
204. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Loading arm assembly. Manually turn the loading arm 360 degrees in a horizontal plane, then vertically move the arm from not less than 30 degrees below, to not less than 70 degrees above the horizontal plane, the drop tube remaining in vertical position during testing. Manually lower the loading arm assembly into its loading position and leave for 2 minutes, then release the loading arm assembly and check its return to original position. Nonconformance to 3.10.1a, b, c, or d shall constitute failure of this test.

4.5.2.2 Static, electrical. Measure the electrical resistance between the tip of the drop tube and the grounding cable attachment clip. Nonconformance to 3.12.1 shall constitute failure of this test.

4.5.2.3 Hydrostatic tests.

4.5.2.3.1 Loading standard. Subject the portion of the loading standard, from the fuel inlet to the loading valve, to a hydrostatic proof pressure of 150 psi, with the loading valve in the closed position. Retain the pressure for a minimum period of 2 minutes, rotate the loading arm 360 degrees and check the loading valve, riser pipe, double swing joint, and O-ring seals for leaks. Check vacuum breaker for proper operation. Nonconformance to 3.1 or 3.10.1.1 shall constitute failure of this test.

4.5.2.3.2 Sliding tube assembly. Subject the sliding tube assembly to a hydrostatic proof pressure of 25 psi and hold for a period of 5 minutes. Nonconformance to 3.10.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.



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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

137. Disassembly not as specified for level A.
138. Unprotected exterior metal surfaces not coated with preservative as specified for level A.
139. Components not lubricated as specified for level A or commercial.
140. Securing and packaging of miscellaneous components not as specified for level A.
141. Box not closed and sealed as specified for level A.
142. Contents not blocked and braced to prevent movement and damage for level A or B.
143. Strapping not as specified for level A.
144. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The loading arm assembly, ground rod, ground wire, and reel shall be disassembled from the riser pipe. Type II loading standards shall have the riser pipe disassembled into its various sections.

5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of components of the loading standards requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Lubrication. Components for which lubrication fittings have been provided shall be lubricated to capacity with the lubricant specified in 3.11.2.

5.1.1.4 Gaskets. Loose gaskets shall be individually preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.5 Anchor bolts, nuts, and washers. The quantity of anchor bolts, nuts, and washers required shall be preserved together in accordance with MIL-P-116, method IC-2, using the contractor's carton or a close-fitting commercial type paperboard or fiberboard box.

5.1.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.



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5.1.1.7 *Miscellaneous components.* The rope or connecting rod shall be secured to the tube of the loading arm assembly. The drop tube shall be folded against and secured to the loading arm. Securing shall be with soft annealed wire or with tape conforming to PPP-T-97, type IV, size 3/4 inch. The valve shall be protected from damage by cushioning with cushioning material conforming to PPP-C-843, type II, class B, secured in place with tape specified herein. The ground wire shall be coiled and the coil secured. The coiled ground wire, reel, grounding rod, technical publications, gaskets, anchor bolts, nuts, and washers and any small loose components removed during disassembly, shall be placed together in a close-fitting box conforming to PPP-B-636, type CF, class weather resistant, variety and grade as applicable to the weight of the contents to be packaged. The contents shall be cushioned to prevent movement or damage. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.2 Commercial. Each complete disassembled loading standard shall be preserved in accordance with ASTM D 3951, and, in addition, components for which lubrication fittings have been provided shall be lubricated to capacity with the lubricant specified in 3.11.2.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The components comprising a complete loading standard, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The contents shall be blocked and braced within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The components comprising a complete loading standard preserved as specified in 5.1, shall be packed as specified in 5.2.1 for level A except the box shall be domestic type or class as applicable. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The components comprising a complete loading standard, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition weight and cube data shall be marked on the shipping container.



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6. NOTES

6.1 Intended use. The stationary loading standards are intended for use in filling of tank railroad cars, tank trucks, and tank trailers.

6.1.1 Type I. The type I rigid loading standard is intended for loading tank trucks, tank trailers, and smaller vehicles and cars.

6.1.2 Type II. The type II sectional loading standard is intended for loading of tank railroad cars, tank trailers, and tank trucks. When this standard is required only for loading of tank trucks and tank trailers, the third (upper) 6-foot section of the riser pipe and the 11-foot long drop tube is not required. For tank railroad cars, the top 6-foot section of riser pipe is used together with the 11-foot length of drop tube. When the intended use condition is not definitely known in advance of procurement, the 6-foot section of riser pipe as well as the 11-foot length of drop tube should be specified.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of loading standard required (see 1.2).
c. When sample units will be required for first article inspection and number of units required (see 3.4).
d. When rope is required in place of connecting rod (see 3.10.1.2).
e. Finish paint if other than as specified (see 3.14).
f. Color of anodic coating (see 3.14.1).
g. When T-deflector is required on drop tube of type I loading standard (see 3.17.1.2).
h. When the third (upper) 6-foot section of riser pipe is not required (see 3.17.2.1).
i. When the 11-foot drop tube is not required (see 3.17.2.2).
j. Degree of preservation and degree of packing required (see 5.1 and 5.2).
k. When other than finish B strapping is required (see 5.2.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a standard production item from the contractor's current inventory (see 4.3). The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

Custodians:
Army - ME
Navy - YD

User activity:
Army - CE

Review activity:
Army - AT

Preparing activity:
Army - ME

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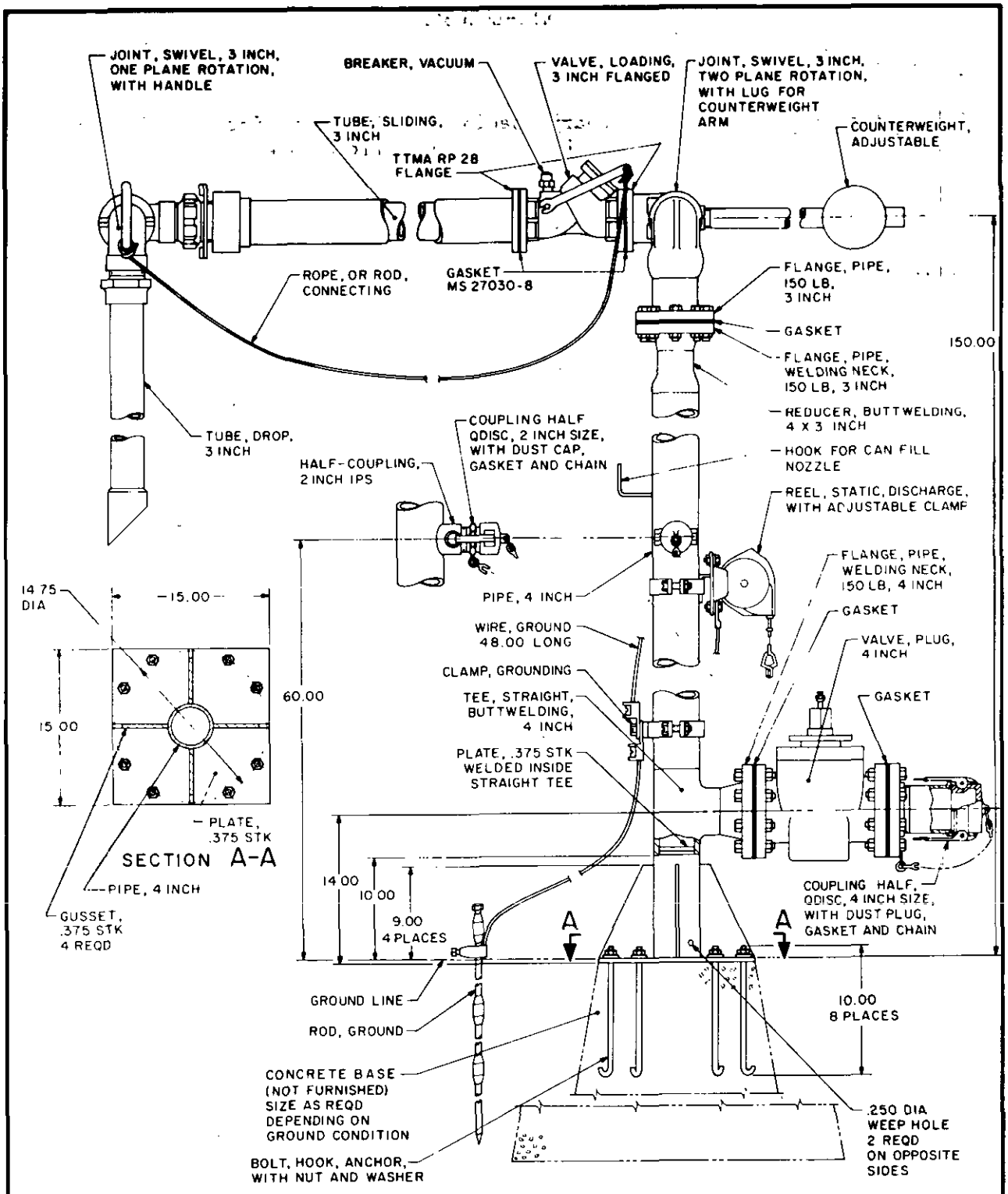



FIGURE 1. Loading standard, liquid transfer, bulk, stationary, type I- rigid.

X-2038B



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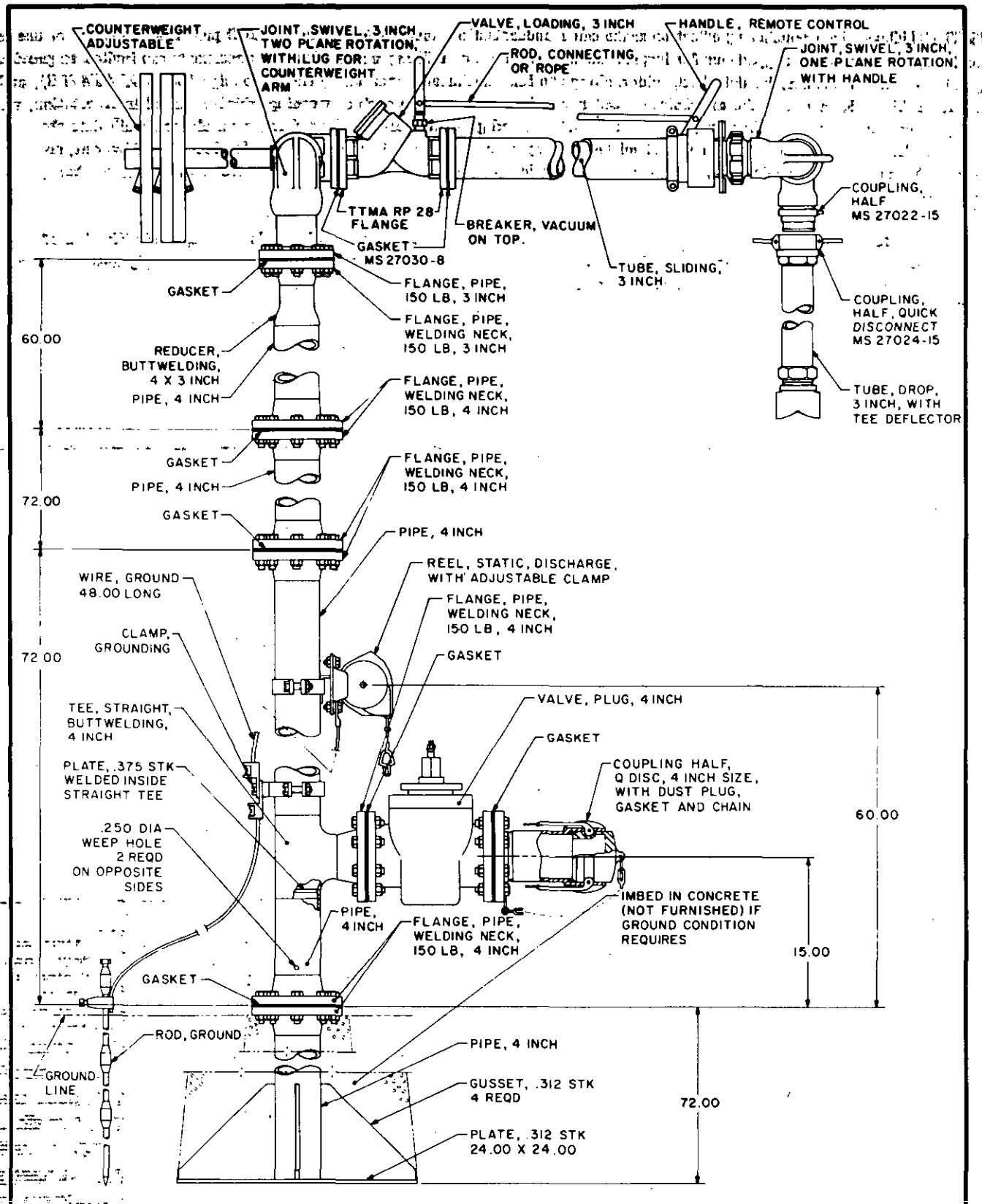



FIGURE 2. Loading standard, liquid transfer, bulk, sta, type II - sectional. X-2039B



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-L-3058F | Loading Standards, Liquid Transfer, Bulk: Liquid Fuels, Stationary, 3-Inch |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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